

Constant rates of change



- 1
- | | | | | | | | |
|---|----------|---|----------|---|----------|---|----------|
| a | Constant | b | Variable | c | Constant | d | Variable |
| e | Variable | f | Variable | | | | |

2 Constant

Instantaneous rates of change

3 0

4 0

5 2

6 2

- 7
- | | | | | | | | |
|---|----|---|---|---|----|---|---|
| a | 0 | b | 0 | c | 2 | d | 1 |
| e | -2 | f | 2 | g | -1 | h | 1 |
| i | -3 | | | | | | |

8 7

9 -6.5

10 2

11 -2

12 $\frac{1}{2}$

13 2

14 -2

a $a = 7$

b $b = 5$



c $c = 2$

16

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	7
1	1.5	0.5	4.75
1	1.1	0.1	3.31
1	1.05	0.05	3.1525
1	1.01	0.01	3.0301
1	1.001	0.001	3.003
1	1.0001	0.0001	3.0003

b 3

17

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	6
1	1.5	0.5	5
1	1.1	0.1	4.2
1	1.05	0.05	4.1
1	1.01	0.01	4.02
1	1.001	0.001	4.002
1	1.0001	0.0001	4.0002

b 4

18



a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
0	1	1	3
0	0.5	0.5	2
0	0.1	0.1	1.487
0	0.05	0.05	1.4355
0	0.01	0.01	1.3959
0	0.001	0.001	0.138 73
0	0.0001	0.0001	1.3864

b 1.386

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a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	-3
1	1.5	0.5	-2.5
1	1.1	0.1	-2.1
1	1.05	0.05	-2.05
1	1.01	0.01	-2.01
1	1.001	0.001	-2.001
1	1.0001	0.0001	-2.0001

b -2

20

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	-2.5
1	1.5	0.5	-3.3333
1	1.1	0.1	-4.5455
1	1.05	0.05	-4.7619
1	1.01	0.01	-4.9505
1	1.001	0.001	-4.995
1	1.0001	0.0001	-4.9995



b -5

Applications

21 -\$10 per $1^\circ F$

22

a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

b -41

c -32

d Linear, because it decreases by the same amount each year.

e Non-linear, because it decreases by more each year.

f Tobias's

g Gwen's

23

a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

b Linear

d 27

e 18

f Mario's

f Christa's



24



a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

- b \$60
- c \$40.8
- d Linear
- e Non-linear
- f Harry's

25

a $-\frac{1}{2}$

- b The amount of water in the bucket decreases by $\frac{1}{2}$ litre every minute.

26

a $\frac{3}{5}$

- b The cost of a call increases by \$0.60 for each additional minute

27 Oliver, 10 km/h

28

- a 0°F per hour b 6°F per hour c -1°F per hour d 1.5°F per hour

29

- a i 9 m/s ii 3 m/s iii -3m/s iv -9m/s

- b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$

- f The ball is falling down.